

An approach for verification of needle cast pathogens with BLAST

The original OTU tables from the PineBiome project have taxonomy assigned to them using the `sklearn-classify` (1) function which implements a naïve-bayes kmer-based approach, in a `qiime2` pipeline (2) on the UNITE (version 9.0) fungal database (3). As the UNITE database in the original pipeline uses dynamic species hypotheses (between 97.5%-99% similarity threshold), it does not contain all species and may limit the discrimination of closely related species such as *Dothistroma pini* and *Dothistroma septosporum*. In addition, it is recommended to verify taxonomic assignment with BLAST (4) where correct assignment of species is important (5,6) and check the ability of the marker sequence to discriminate at the species levels (7). To address these issues we follow a five-step process (Fig 1) that incorporates extraction of potential needle-cast OTU sequences, BLAST assignment against UNITE (version 10.0) 99% SH database and the INSD database (8) restricted to fungal hits; then assessment of taxonomic conflict between the approaches. R (version 4.4.2)(9) was used with the packages: `tidyverse-2.0.0` (10), `qiime2R-0.99.6` (11), `phyloseq-1.50.0` (12), `microviz-0.12.5` (13), `microbiome-1.28.0` (14), `xml2-1.3.8` (15), `rgbif-3.8.4` (16), `rvest-1.0.4` (17), `DECIPHER-3.2.0` (18) for all other processing of the data.

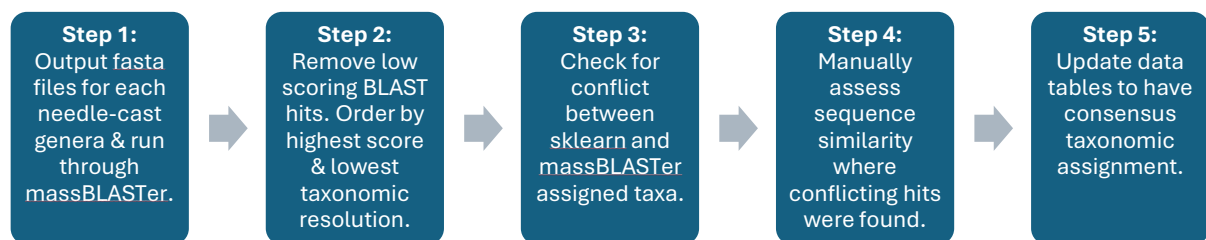


Figure 1. Overview of the steps used to verify the identify of needle cast pathogens.

A total of 10 genus were apriori identified as containing potential needle cast pathogens: *Dothistroma*, *Lophodermium*, *Lophodermella*, *Coleosporium*, *Cyclaneusma*, *Gremmeniella*, *Gremmenia*, *Pachyramichloridium*, *Rhizosphaera*, *Sirococcus* (??REF??). If these were present in the dataset then the fasta files were extracted and run through massBLASTer (BLASTN 2.15.0+) regular mode in the plutoF platform (19). Html of the BLAST results were imported into R and processed to produce data tables of hits. RGBIF was used to fill in missing taxonomic levels. Those that failed to meet the thresholds for suitable blast assignment (query length > 100bp, query coverage >80%, E value < 1×10^{-20} , Percentage similarity > 97.5%) were removed and the remaining hits ordered by highest score and then greatest taxonomic resolution (i.e. two hits with the exact same score and one is classified to species level and the other to genus level the species level assignment will be rank higher). The top ranking hit for each needle cast OTU was then chosen to represent the BLAST assignment.

Taxonomic conflicts were identified by assessing for matches between the naïve bayes and BLAST-classified taxonomies from kingdom to species level. If a conflict was identified at any level the OTU was indicated as requiring manual verification. Manual verification steps (Fig 2.) of synonym checking, comparison of ITS sequences of the species and incorporation of any prior knowledge of the taxa associated with the study system were performed. Consensus assignments were then integrated back into the data frames prior to calculation of the relative abundance of needle cast pathogens.

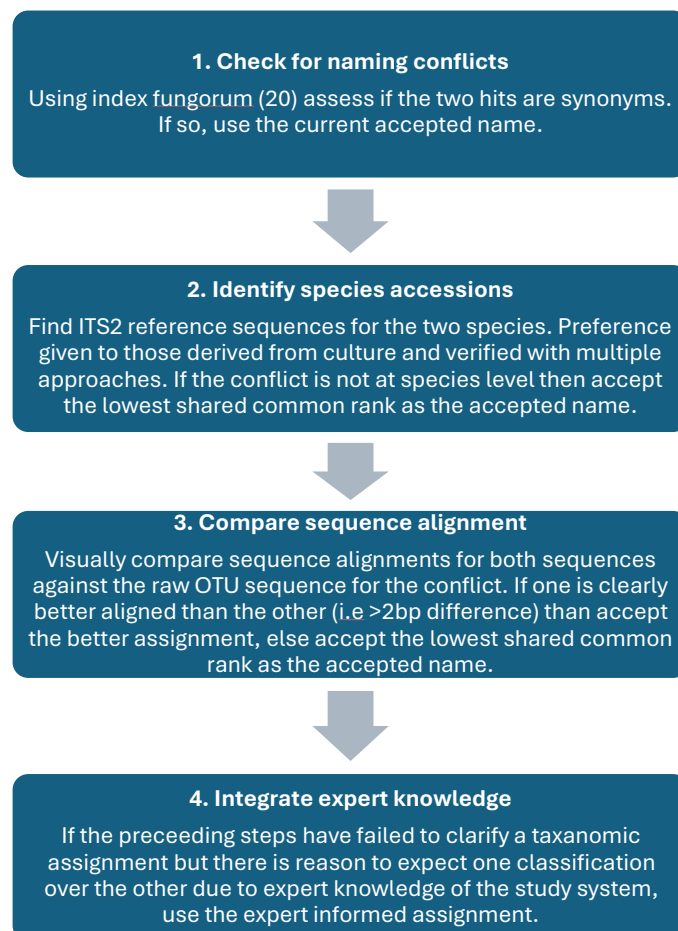


Figure 2. Manual verification steps used to compare conflicting taxonomic assignments. Accession number, references for accession numbers and justification for accepted classification as all recorded as appropriate throughout the manual verification steps.

Data & code availability

Fasta files for needle cast genera, massBLASTER html output files, manual verification notes and R script are all available in

https://github.com/aphidbeth/DNB_Survey_Paper_OTU_processing.

References

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